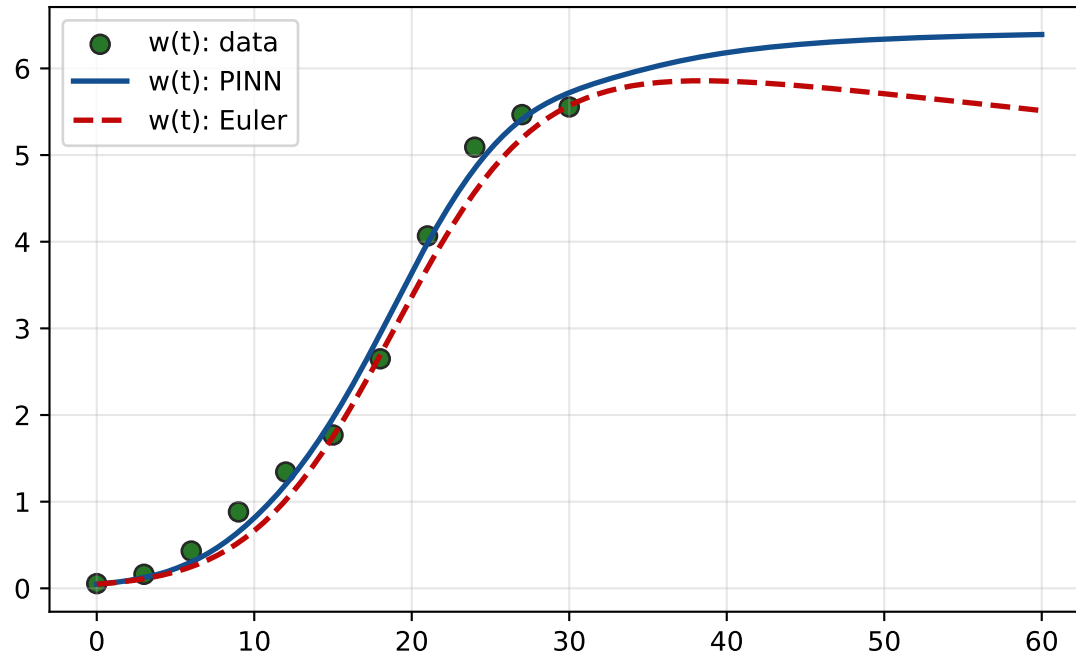
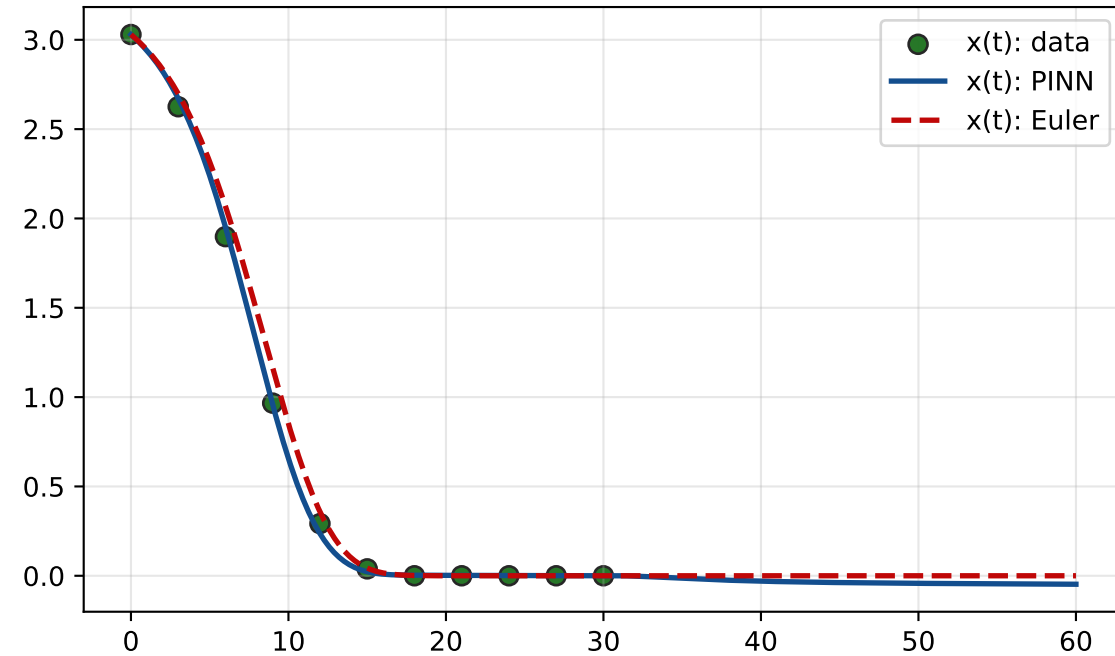


# E1 | PINN vs Euler

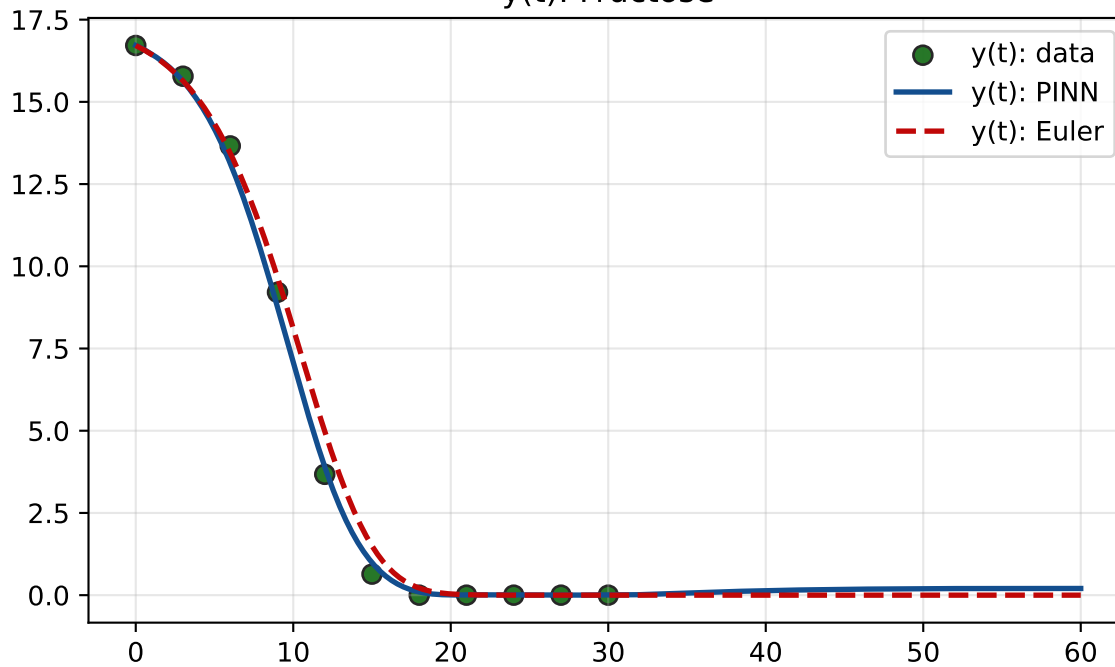
w(t): Biomass



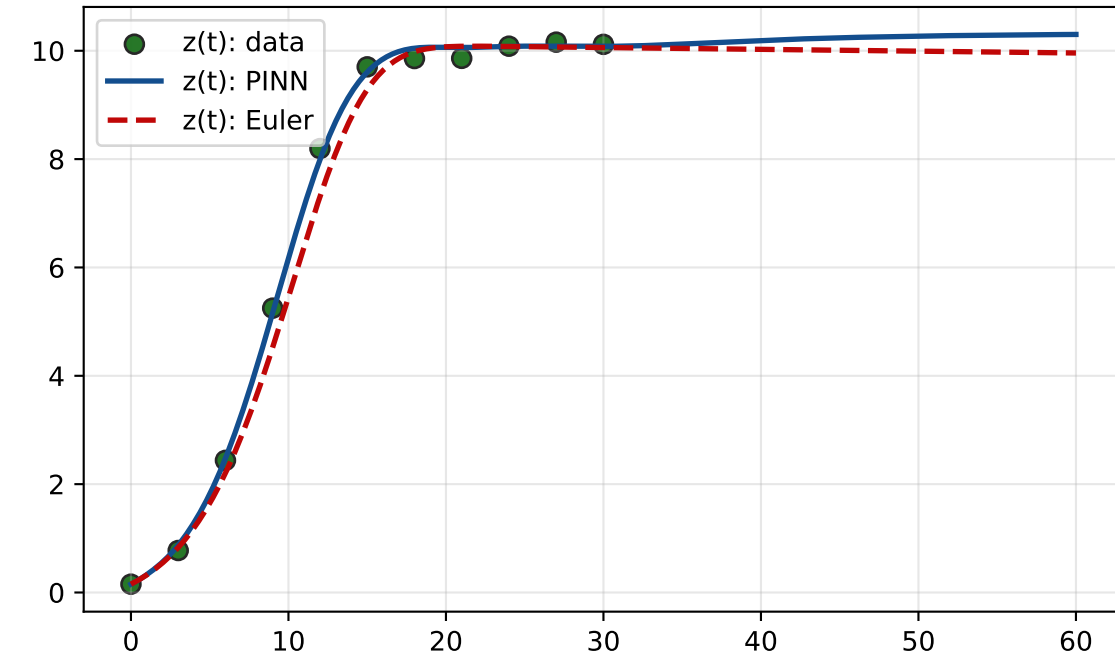
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E1

R2=0.9927, R2\_adj=0.9916, RSS=5.189e+00, AIC=-7.978e+01

cond(JTJ)=9.82e+03, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 2.150662e-01 | 5.251877e-02 | 1.063187e-01 | 1.087475e-01  | 3.213849e-01 | 2.127857e-04 |
| p2    | 3.514421e-02 | 1.052963e-02 | 2.131612e-02 | 1.382809e-02  | 5.646033e-02 | 1.899401e-03 |
| p3    | 7.493263e-02 | 8.510204e-02 | 1.722801e-01 | -9.734746e-02 | 2.472127e-01 | 3.841218e-01 |
| p5    | 5.186478e-01 | 1.389604e-01 | 2.813106e-01 | 2.373372e-01  | 7.999585e-01 | 6.193116e-04 |
| p7    | 2.931570e-01 | 4.687807e-02 | 9.489969e-02 | 1.982573e-01  | 3.880567e-01 | 2.556519e-07 |
| p9    | 1.592830e-01 | 2.527419e-02 | 5.116493e-02 | 1.081181e-01  | 2.104479e-01 | 2.193363e-07 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 4.709665e-02 |
| x0 (fixed)     | 3.029527e+00 |
| y0 (fixed)     | 1.671414e+01 |
| z0 (fixed)     | 1.538119e-01 |
| W0 (fixed)     | 0.000000e+00 |